From Reversible Bit-Rings to Quantitative Matter Readouts

The Qu Relational Compiler, Three-Dimensional Steric Closure, and Symmetry-Reduced Polymer Mechanics

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1. The Architectural Shift: From State Catalogs to Relational Compilation

A foundational problem across discrete mathematics, quantum simulation, and macromolecular physics is distinguishing between a static state catalog and an executable transformation architecture. Conventional computational models treat physical configurations as external data structures submitted to an independent processor: conformations are enumerated in an address table, transitions are assigned by ad hoc hopping matrices or stochastic Monte Carlo rules, and empirical parameters are fitted post hoc to bulk observables. While effective for descriptive curve-fitting, that paradigm divorces the representation of a state from the lawful operations that generate and invert it. Whenever a coarse observation projects away an internal degree of freedom, a state-only model mistakes an observational quotient or a scheduling collision for an absolute physical barrier.

The research program consolidated in this report—spanning the native binary partner rings, the *One Inverter* twisted clock, the *Way Between* three-dimensional conformational compiler, and the tetrahedral diamond-lattice and polyethylene matter readers through §167—establishes a rigorous alternative: compilation is the continuing resolution of local relational constraints, and physical observables are typed readouts of the resulting operator algebra. Rather than postulating an external machine or assigning arbitrary binary labels to particles, the architecture begins with the minimal condition for a distinction to persist without terminal fixed points (C_0/C_1) and traces the exact algebraic and geometric structures forced at each scale. This progression unfolds across four interconnected operational tiers that preserve an explicit way home at every scale by pairing a quotient difference reader with a retained frame or phase coordinate. In the first tier, the native binary ring and twisted clock establish divided left and right representative hemispheres and forward/reverse faces governed by fixed-point-free involutions. In the second tier, the planar honeycomb

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route compiler lifts into a three-dimensional control torus where synchronous bond torsions execute a diagonal bypass around blocked planar corners. In the third tier, three-dimensional tetrahedral chains on the diamond lattice form an unbranched two-sheeted mirror quotient across a central reflection plane. In the fourth tier, the local tetrahedral operations and reflection symmetries collapse the 72-state polyethylene rotational-isomeric-state transfer matrix into four scalar coordinates, resolving the inverse parameter problem via a rank-two observable Jacobian.

Across all four tiers, the compiler operates through two governing equations that separate the underlying accumulated transformation from the perspective of a particular observer:

$$\boxed{G_n = G_0 g_1 g_2 \cdots g_n} \qquad \text{and} \qquad \boxed{r_H(W) = \langle m(W), f_H \rangle}$$

In the first equation, local reversible increments g_j—whether bit shifts, modular carries, planar turns, or spatial $SE(3)$ bond torsions—accumulate relative to a held reference frame G_0. Changing G_0 reorients the global description rigidly without altering any internal relation g_j, while substituting a subword $g_{a+1} \cdots g_b$ with an alternative sequence $g'_{a+1} \cdots g'_b$ having the identical running product produces a strictly local transformation whose exterior remains undisturbed. In the second equation, the multiplicity structure $m(W)$ belongs to the object's own relation complex, while a chosen reader H exposes only the projection f_H. An apparent zero in $r_H(W)$ or a rank deficiency in a planar Jacobian does not imply the absence of a physical continuation; it diagnoses a specific mismatch between the active transformation and the reader's coordinate chart.

Every result included in this synthesis adheres to the strict execution discipline established across the project audit trail. Claims are partitioned into four explicit epistemic grades: STANDS for proven mathematical theorems and established identities; RUNS for exact finite-domain or symbolic executions verified by self-contained code; DEAD END for falsified hypotheses whose true structural owners have been identified and logged; and OPEN for prospective links requiring independent empirical or analytical closure.

2. Native Partner Rings, the Twisted Clock, and Cross-Width Transport

2.1 The Fixed-Point-Free Partner Involution and Canonical Addressing

At the base of the discrete compiler is an n-bit word space $\{0,1\}^n$ of size $N = 2^n$ equipped with two elementary involutions: bit reversal $J_n(b_0 b_1 \cdots b_{n-1}) = b_{n-1} \cdots b_1 b_0$ and bitwise complementation $C_n(x) = x \oplus (2^n - 1)$. While C_n is fixed-point-free at every width, J_n fixes all $2^{\lceil n/2 \rceil}$ palindromes ($J_n(x) = x$). To prevent a state from partnering with itself—a twin collision that would stall continuation in a reversible circuit—the native partner operation P_n substitutes complementation exclusively on the palindromic branch:

$$\boxed{P_n(x) = \begin{cases} J_n(x), & J_n(x) \neq x, \\ C_n(x), & J_n(x) = x. \end{cases}}$$

Because J_n and C_n commute and C_n preserves palindromicity, P_n is a fixed-point-free involution ($P_n^2 = I$, $P_n(x) \neq x$) that commutes with complementation ($P_n C_n = C_n P_n$) across all widths $n \geq 1$. The 2^n words partition into $M = 2^{n-1}$ disjoint partner pairs $\{a, P_n(a)\}$, each represented canonically by its smaller numerical member $a = \min(x, P_n(x))$. The addressable ring R_n places the M canonical representatives in ascending order on the left hemisphere ($0 \leq i < M$) and their partners in descending order on the right

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hemisphere ($M \le i < N$), followed by a single center swap of the equatorial seam indices $M-1$ and M. This construction enforces the universal facing law at every index $i \in \{0, \dots, N-1\}$:

$$\boxed{R_n[2^n - 1 - i] = P_n(R_n[i]).}$$

A critical correction to simplified descriptions of R_n is that canonical representatives $a < P_n(a)$ do not all begin with most-significant bit (MSB) zero. Whenever a word has outer frame $(b_0, b_{n-1}) = (1,1)$ and a non-palindromic interior $w \in \{0,1\}^{n-2}$ ($J_{n-2}(w) \neq w$), both $x = 1w1$ and its partner $P_n(x) = 1J_{n-2}(w)1$ begin with 1, whereas their complements $0C_{n-2}(w)0$ and $0J_{n-2}(C_{n-2}(w))0$ both begin with 0. The exact count of MSB-1 canonical representatives at width $n \ge 3$ is:

$$\boxed{N_{\text{MSB=1}}(n) = 2^{n-3} - 2^{\lfloor (n-3)/2 \rfloor},}$$

yielding 0,1,2,6,12,28,56,120 for $n = 3 \dots 10$. Correspondingly, the exact rank $r_n(p)$ of canonical representatives below threshold p requires counting all palindromes $\pi_n(p)$ and clamping canonical palindromes at $T_n(p) = \min(\pi_n(p), 2^{\lceil n/2 \rceil - 1})$ alongside the comparator sum $\Delta_n(p) = \sum_{x<p} \operatorname{sgn}(J_n(x) - x)$:

$$\boxed{r_n(p) = \frac{p - \pi_n(p) + \Delta_n(p)}{2} + T_n(p).}$$

Combined with the two-step E/G cross-width lift—which inserts a middle bit $I_m(x) = x_{0:h}\, m\, x_{h:L}$ to reach odd widths and branches into $E_m(x) = \text{canon}(mxm)$ and $G_m(x) = 0P_L^m(x)1$ to reach the next even width—the rank and unrank operators resolve any word's physical ring address in $O(n)$ bit operations without materializing the 2^n-element array, as verified across 2,097,148 thresholds for $n \le 20$ and scalar queries up to $n = 1{,}024$.

2.2 The Emergence of the Klein Four-Group at Width Three, the Twisted Clock, and Odd-Lane Locks

Evaluating the operator group generated by $\{I, P_n, C_n, P_nC_n\}$ reveals a sharp structural transition at width three. For $n = 1$ and $n = 2$, every word is either a palindrome or an anti-palindrome, forcing $P_n = C_n$ so the operator group has order 2. At $n = 3$ ($N = 8$), the first non-palindromic, non-antipalindromic Klein 4-orbit $\mathcal{O}_{001} = \{001,100,110,011\}$ appears, separating $P_3 \neq C_3$ into the four-element Klein group $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. On the three nontrivial Walsh character modes of $\mathcal{O}_{001}$, the operations P, C, PC act as the three diagonal half-turn matrices $\text{diag}(1,-1,-1)$, $\text{diag}(-1,1,-1)$, and $\text{diag}(-1,-1,1)$ in $SO(3)$ ($\det = +1$), while a faithful two-dimensional reflection representation $\text{diag}(-1,1)$, $\text{diag}(1,-1)$, $\text{diag}(-1,-1)$ in $O(2)$ also exists. Simultaneously, the marginal 1-bit, 2-bit, and 3-bit readers at $n = 3$ form a cellular chain complex over $\mathbb{F}_2$ with $V = 6$ poles, $E = 12$ arcs, and $F = 8$ octant faces ($\operatorname{rank}\partial_1 = 5$, $\operatorname{rank}\partial_2 = 7$, Betti numbers $b_0 = 1, b_1 = 0, b_2 = 1$, and Euler characteristic $\chi = V - E + F = 2$), whose dual adjacency graph is the three-bit cube Q_3.

In the *One Inverter* derivation (§143, seal 00981594…), applying the single-bit continuation law (only NOT is fixed-point-free on $\{0,1\}$) to an n-bit shift register produces the affine twisted rotation τ:

$$\boxed{\tau(x_0x_1 \cdots x_{n-1}) = x_1 \cdots x_{n-1}\neg x_0, \qquad \tau^k(x) = \text{rotl}_k(x) \oplus (2^k - 1) \quad (0 \le k \le n).}$$

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After n ticks, every bit has passed through the single inverter once, yielding the global complement ($\tau^n = C_n$), and after $2n$ ticks the state returns ($\tau^{2n} = I$), partitioning $\{0,1\}^n$ into orbits of lengths dividing $2n$ whose total count is OEIS A000016. At $n = 2$, τ_2 is the Gray quarter-turn $00 \to 01 \to 11 \to 10$, uniting the temporal half-step and spatial quarter-turn into one operator. Furthermore, the inward E/G route terminates at the central two-bit XOR reader $r_n(x) = x_{n/2-1} \oplus x_{n/2}$ (the innermost shell lamp of $x \oplus J_n(x)$). Under τ, the temporal trace $r_n(\tau^t x)$ reads each adjacent bit difference around the cyclic word with one inverted seam crossing, satisfying $\bigoplus_{t=0}^{n-1} r_n(\tau^t x) = 1$ for even n and recovering the full n-bit word from $n-1$ temporal reads plus one canonical face bit.

When $n = qs$ for an odd number of lanes q, the alternating-complement lift $\Lambda_q(u) = u \parallel C_s(u) \parallel u \parallel \cdots \parallel u$ embeds an s-bit compiler directly inside the qs-bit compiler, intertwining the twisted clock, partner map, and central reader:

$$\boxed{\tau_{qs}\Lambda_q = \Lambda_q\tau_s, \qquad P_{qs}\Lambda_q = \Lambda_q P_s, \qquad r_{qs}(\Lambda_q(u)) = r_s(u), \qquad d_H(\Lambda_q u, \Lambda_q v) = q \cdot d_H(u,v).}$$

No solution to $\tau^s x = C_n x$ exists for even q, and no two lifted words $\Lambda_q(u), \Lambda_q(v)$ are ever adjacent under the global ring successor T.

2.3 Traversal Orthogonality, the Square-Root Bridge, and All-Width Branch Transport

Comparing the native ring successor $T(R_n[i]) = R_n[(i+1) \bmod N]$ against its complement-conjugate traversal $T_C = C_n T C_n$ exposes a universal operator-space geometry. By the facing law, $P_n T P_n = T^{-1}$, which implies $T_C = (P_n C_n) T^{-1} (P_n C_n)$. In permutation-matrix space, the Frobenius inner product $\langle T, T_C \rangle_F = \mathrm{tr}(T^{\mathsf{T}} T_C) = \#\{x : T(x) = T_C(x)\}$ vanishes identically across all tested widths $n = 2 \ldots 22$ ($\| T - T_C \|_F^2 = \| T \|_F^2 + \| T_C \|_F^2 = 2^{n+1}$, an exact 90° angle), and the output difference mask $\Delta(x) = T(x) \oplus T_C(x)$ is strictly complement-invariant ($\Delta(C_n x) = \Delta(x)$).

The commutator bridge $B = T_C T^{-1}$ carries each T-destination directly to the corresponding T_C-destination ($B(T(x)) = T_C(x)$) and factors algebraically as the exact square of a half-step operator S:

$$\boxed{B = T_C T^{-1} = S^2, \qquad S = P_n C_n T^{-1} = C_n \circ \tau_{\text{addr}},}$$

where $\tau_{\text{addr}}(R_n[i]) = R_n[(-i) \bmod N]$ is the ring-index reflection fixing only the Origin $R_n[0] = 0$ and the Seam $R_n[N/2]$.

Width n	Ring Size N=2n	Frobenius ⟨T,TC⟩F	S-Cycle Lengths	B=S2 Cycle Lengths	Spinal Fraction	Prism Components (B+M)	Bipartite Graph?
2	4	0	[4]	[2,2]	100.0%	1	True

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Width n	Ring Size N=2n	Frobenius ⟨T,TC⟩F	S-Cycle Lengths	B=S2 Cycle Lengths	Spinal Fraction	Prism Components (B+M)	Bipartite Graph?
3	8	0	[8]	[4,4]	100.0%	1 (Q_3)	True
4	16	0	[16]	[8,8]	100.0%	1 ($C_8 \square K_2$)	True
5	32	0	[32]	[16,16]	100.0%	1 ($C_{16} \square K_2$)	True
6	64	0	[32,16,16]	[16,16,8,8,8,8]	50.0%	3	True
7	128	0	[96,16,16]	[48,48,8,8,8,8]	75.0%	3	True
8	256	0	[170,27,27,16	[85,85,27,27,8,8,8,8]	66.4%	4	**False** (odd 27,85)
9	512	0	[466,16,16,7,7	[233,233,8,8,8,8,7,7]	91.0%	4	**False** (odd 7,233)
10	1,024	0	[936,28,28,16	[468,468,14,14,14,14,	91.4%	5	True

As recorded in the cycle census, the alternating $(C_n, \tau_{\text{addr}})$ spine connects Origin to Seam across 100% of the ring for $n = 2 \ldots 5$, but detaches at $n = 6$ ($N = 64$) into a 32-state core and two 16-state branches. Because $P_n C_n$ preserves the left representative hemisphere $\text{Rep}_n = \{x : x < P_n(x)\}$ and $T = S^{-1} \circ P_n C_n$, every detached S-orbit is trapped 100% inside Rep_n or 100% inside Part_n, occurring in complementary mirror pairs $(U, C_n(U))$ with zero direct T-edges between opposite hemispheres and generating the

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symmetric T-transition matrix $\begin{pmatrix} 22 & 5 & 5 \\ 5 & 11 & 0 \\ 5 & 0 & 11 \end{pmatrix}$ at $n = 6$. Pairing each B-cycle A with its disjoint complement $C_n(A)$ via $M(B^j a) = B^j q_0$ (with $q_0 = S^{-1}(0)$ on the origin pair and $q_0 = C_n(a)$ on detached pairs, seal 39c153fa…) constructs a fixed-point-free involution M satisfying $M^2 = I, MB = BM, MC = CM$ across all $131{,}068$ states for $n = 2 \ldots 16$, even when odd B-cycles appear at $n = 8{,}9{,}14{,}15$.

Moreover, the detached branches obey all-width symbolic predecessor certificates (seals d8221144… and 250ecd1b…). The 16-state branch is generated for every $n \geq 6$ by repeating a single middle bit $m_t \in \{0{,}1\}$ between 3-bit headers and footers ($x_t^{(n)} = u_t\, m_t^{n-6}\, v_t$, having 11 internal and 5 external T-ports at $n = 6$ and locking at 4 internal and 12 external T-ports for all $n \geq 7$). Similarly, the 28-state left branch at $n = 10$ (anchored at seed 23) lifts to all $n \geq 10$ via a 5-bit header/footer template $x_t^{(n)} = u_t\, m_t^{n-10}\, v_t$ with 28-bit middle-phase word 0101010101010110101010101010, predecessor decrements $d \in \{1{,}2{,}4\}$ (borrowing across the middle run only at phase 27), 1 internal and 27 external T-ports, and closed rung phase $M(x_t) = C(x_{-t})$ for even t and $C(x_{6-t})$ for odd $t \pmod{28}$, verified through $n = 2{,}048$.

3. Dual-Read Stream Injection and Exact Address Jumps in Sixty-Four-Bit Rings

3.1 The Center-Two Quotient and Symmetric Partner-Fold Residues

At $n = 64$ ($N = 2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$ words), the symbolic rank/unrank compiler operates directly on 64-bit streams without memory materialization (seals 081ed4ff… and 222e1249…). For any even width $n \geq 4$, removing the two central bits center2$(x) = (y, m_{\text{mid}})$ commutes with the partner map ($\pi(P_n(x)) = P_{n-2}(y)$) everywhere except when the $(n-2)$-bit core y is a palindrome ($J_{n-2}(y) = y$) and the central pair has odd parity ($m_{\text{mid}} \in \{01_2, 10_2\}$), in which case $\pi(P_n(x)) = y$ instead of $C_{n-2}(y)$. Across the full ring R_n, there are exactly $2^{n/2}$ such exceptions ($2^{32} = 4{,}294{,}967{,}296$ words in R_{64}), and retaining the 1-bit XOR parity of the central pair selects the exact return branch.

Injecting two ASCII byte streams—Ovaltine (b"Dont forget to drink your Ovaltine", 34 bytes, 105 reads at a 2-bit stride) and Singularity (b"this is it. this is the Singularity.", 36 bytes, 113 reads)—into a 64-bit aperture demonstrates that while the raw input words and ring addresses fluctuate irregularly, the partner-fold difference $D(x) = x \oplus P_{64}(x)$ is identically a nonzero palindrome ($J_{64}(D(x)) = D(x) \neq 0$) on every read, locking the cumulative bit-lamp activity $L(k) = L(63-k)$ symmetrically across all 32 concentric shells.

3.2 Tagged Two-Bit Shifts, Priority-Encoder Boundaries, and the Parabola Staircase

Advancing the 64-bit aperture by two bits transforms $x = a \parallel y$ (where $a \in \{0{,}1\}^2$ is the departing prefix and $y \in \{0{,}1\}^{62}$ is the retained body) into $x_b = y \parallel b = (4x + b) \bmod 2^{64}$. Retaining the departing pair a provides the exact inverse $(x_b, a) \mapsto (a \parallel (x_b \gg 2),\ x_b \& 3)$, while bit reversal intertwines the forward shift with a reverse shift: $J_{64}(y \parallel b) = J_2(b) \parallel J_{62}(y)$. Consequently, when the reverse rail is canonical ($J_{64}(x_b) < x_b$), a 1-bit change in the incoming LSB pair b flips the MSB of the reversed word, redirecting the canonical ring address $u(x) = (r(c(x)) + s(c(x)))/2^{64}$ across the ring.

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Stream & Dual-Read Metric (R64R_{64}R64, 2-Bit Stride)	**Ovaltine Stream (105105105 Reads, 104104104 Jumps)**	**Singularity Stream (113113113 Reads, 112112112 Jumps)**	**Combined / Theoretical Compiler Invariant**
Left / Right Hemisphere Visits	54 Left / 51 Right	57 Left / 56 Right	Exact global balance (2^{63} pairs)
Face-Selection Prefix Gate (forward / tie / reverse)	38 / 35 / 31	42 / 34 / 36	$147/216$ (68.1%) resolved at outer 2-bit gate
Mean Outer Face-Deciding Shell k_{out}	1.495238	1.353982	First lit mismatch $g_k = b_k \oplus b_{63-k} = 1$
Mean Inner Quotient-Boundary Shell k_{in}	29.590476	29.752212	Last lit mismatch $g_k = 1$ before palindromic core
Jumps Altered Relative to Appending oo	$83/104$	$86/112$	$169/216$ (78.2%) input-redirected
Mean / Max ‖ Δu_{rep}	0.162913 / 0.360830	0.155280 / 0.359919	Exact $\Delta u = \Delta f + \Delta\varepsilon$ along $f(t) = t - t^2/2$
Scaled Parabola Error $\|\varepsilon(c)\| \cdot 2^{3}$ (Min / Max)	0.008168 / 0.247958	0.008699 / 0.247825	Asymptotic palindrome constant $\approx 0.2453 < 0.25$
Antipodal Partner-Jump Cancellation $\Delta i(Px) = -\Delta i(x)$	$104/104$ PASS	$112/112$ PASS	$216/216$ exact; $0/216$ match native T-steps

In Singularity step 0 (7468697320697320), shifting with b = oo lands at canonical coordinate $u = 0.004683294804$ ($\Delta u = -0.013918$), whereas shifting with actual input b = o1 flips the reversed MSB

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(0133a58133a5858b → 8133a58133a5858b), jumping across the 11-frame seam to $u = 0.377336138315$ ($\Delta u = +0.358735$). Across all 216 transitions, the opposed-bit differences $g_k = b_k \oplus b_{63-k}$ unify the two ends of the shell hierarchy: the outermost lit shell $k_{\text{out}} = \min\{k: g_k = 1\}$ decides the canonical orientation $x \lessgtr J(x)$, while the innermost lit shell $k_{\text{in}} = \max\{k: g_k = 1\}$ locates the boundary where the inner core becomes palindromic.

4. The Planar Honeycomb Compiler, Parity Sectors, and Precondition Gates

4.1 Padding Leakage in Address-Catalog Hopping versus Direct Route Compilation

The necessity of compiling molecular geometry directly from turn words is established by the §144 audit of the six-residue cyclic PHPPHP square-lattice model (seal of6ocd3e…). Encoding the 6 valid 2×1 domino conformations into an 8-state 3-qubit register ($R_3 = [0,1,2,6,3,5,4,7]$) forces two register addresses ($3 = 110_2$ and $7 = 111_2$) to represent invalid padding states. Pairwise coordinate comparison of the 15 conformation pairs proves that zero pairs differ in 1 or 2 residue positions (6 pairs differ at 3 sites and 9 pairs differ at 4 sites). Yet imposing a cyclic address-hopping kinetic term $K = -0.7(T_8 + T_8^{\dagger})$ introduces four edges crossing the valid/invalid boundary—(2,3), (3,4), (6,7), (7,0)—so $[H, P_{\text{valid}}] \neq 0$ ($\| QKP \|_2 = \sqrt{2}(0.7) \approx 0.989949$). From a uniform valid initial state, invalid-state probability grows at $\frac{49}{75}t^2 + O(t^3) \approx 0.653333t^2$ regardless of any finite energy penalty Λ (leaking $0.324402, 0.006480, 0.000152, 0.000024$ at $t = 1$ for $\Lambda = 0, 10, 50, 100$), while Schur-complement elimination $H_{\text{eff}}(E) = H_{PP} + H_{PQ}(E - \Lambda I_2)^{-1} H_{QP}$ reveals that the padding states act as artificial virtual bridges of amplitude $-0.49/\Lambda$ between (0,6) and (2,4). Deleting those unphysical bridges via $K_{\text{valid}} = PKP$ eliminates leakage entirely but severs the valid space into two disconnected three-state paths (0–1–2 and 4–5–6), bounding hydrophobic contact probability by $p_{\text{contact}}(t) \leq 1/2$ (0.463631 at $t = 1$).

To eliminate catalog padding, the honeycomb compiler (§§145–155, seals a88c0432…, daac3a89…) encodes an $(m + 2)$-bead chain directly as an m-bit turn word $x = b_0 \cdots b_{m-1} \in \{0,1\}^m$, where the first bond points east from (0,0) to (1,0) and each bit $b_i \in \{0,1\}$ rotates the heading by $+60°$ or $-60°$ across the six unit directions $D = ((1,0), (0,1), (-1,1), (-1,0), (0,-1), (1,-1))$. The legal domain $V_m = \{x \in \{0,1\}^m : |\{p_0(x), \dots, p_{m+1}(x)\}| = m + 2\}$ contains 2,4,8,16,30,58,112,216,406,776,1472,2796,5260,9964 self-avoiding conformations for $m = 1 \dots 14$ and reaches 19,053,856 conformations at $m = 26$.

4.2 Two Readings of One Word, Mirror Parity Sectors, and the Precondition-Gate Hierarchy

On the honeycomb lattice, every turn word x carries two complementary readings that link directly back to *One Inverter* (§154, §162, seal a185394f…). First, the intrinsic shape reading is captured by the $(m - 1)$-bit adjacent-difference word $\delta(x)_i = b_{i-1} \oplus b_i$, which records whether the internal dihedral at bond i is cis (0°, $\delta_i = 0$) or trans (180°, $\delta_i = 1$). Across all 8,188 words for $m = 2 \dots 12$, two chains have identical ordered pairwise-distance matrices if and only if their difference words $\delta(x)$ coincide. Second, the frame reading resolves the kernel of δ, which is the two-element complement orbit $\{x, C_m(x)\}$. In integer lattice coordinates, $C_m(x)$ sends every vertex $(a, b) \mapsto (a + b, -b)$, which in three-dimensional Euclidean space $(a + b/2, \frac{\sqrt{3}}{2}b, 0)$ is the proper 180° spatial rotation $R_{\pi,x}$ about the first bond axis. Retaining the single frame bit $f = b_0$ (or canonical ring face face$(Cx) = 1 -$ face(x)) reconstructs x bijectively via $b_i = f \oplus$

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$\oplus_{j=1}^{i}\delta_j$. Furthermore, the native ring partner $P_m(x)$ satisfies $\delta(P_m x) = \mathrm{rev}(\delta(x))$ identically across all widths, corresponding to reading the unlabeled spatial chain from its opposite end.

Because C_m preserves self-avoidance and all hydrophobic-polar (E_{HP}) contact counts (0 of 776 energies changed by C_{10} on HHPHPPHHPPHP at $m = 10$, whereas J_{10} and P_{10} alter 48 of 776 due to sequence reversal), any C_m-equivariant move adjacency $A = A_0 + A_1$ commutes with $U_C|x\rangle = |C_m x\rangle$. A two-point Fourier transform $|a, \pm\rangle = (|a\rangle \pm |C_m a\rangle)/\sqrt{2}$ block-diagonalizes $H = V - \gamma A$ into exact even and odd parity sectors $H_\pm = V_q - \gamma(A_0 \pm A_1)$, cutting the $m = 10$ state vector from 776 to 388 complex amplitudes and agreeing with full-space propagation to machine precision.

As the move vocabulary expands from single-bit collective pivots (120° tail rotations) to bounded-support local rewrites, the compiler exposes a strict separation between GF(2) mask span, substring precondition gates, and steric self-avoidance:

Move Vocabulary	**Active Masks / Generators**	**Reaches Mirror C(x)C(x)C(x) (1m∈Span1^m \in \text{Span} 1m∈Span)**	**m=10m=10m=10 Edges**	**m=10m=10m=10 Comps**	**m=10m=10m=10 Contact Reach (/776/776/776)**	**m=6…14m=6 \dots 14m=6…14 Even Component Counts**
pivot (collective)	All m 1-bit flips e_j	All $m \geq 1$	3,352	1	776	1,1,1,1,1 (connected $m = 2 \ldots 14$)
tail4 ($\leq$ 4-bead tail)	e_{m-4}, e_{m-3}, e_m	$m \leq 4$ only	1,374	58	88	Center bit frozen for $k < m/2$
hex (2-bead local)	1001 ↔ 0110 (1111)	$m \equiv 0 \pmod 4$	396	404	42	30,112,404,14
hybrid (hex + e_{m-1})	hex + last bit e_{m-1}	$m \equiv 0,1 \pmod 4$	768	192	44	Bead $m-1$ frozen (69 sites at $m = 10$)

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Move Vocabulary	**Active Masks / Generators**	**Reaches Mirror C(x)C(x)C(x) (1m∈Span1^m \in \text{Span} 1m∈Span)**	**m=10m=10m=10 Edges**	**m=10m=10m=10 Comps**	**m=10m=10m=10 Contact Reach (/776/776/776)**	**m=6…14m=6 \dots 14m=6…14 Even Component Counts**
back2 (hex + last 2)	hex + e_{m-2}, e_{m-1}	$m \not\equiv 3 \pmod 4$	1,118	73	64	Max eig diff 0.328473 at $m = 10$
both2 (hex + both 2)	hex + e_0, e_1, e_{m-2}, e_m	**All** $m \geq 4$ (to $m \leq 257$)	1,840	14	612	2,5,14,35,106 (traps $\mathcal{Z}_m$)
both2 + §148 10-Cycle	both2 + 5-edge 10-cycle ($i \geq 1$)	All $m \geq 4$	2,240	3	776	1,3,3,5,17 (exits $\mathcal{Z}_m$ at $i = m - 4$)
both2 + Sym 10-Cycle	both2 + 10-cycle ($i \geq 0$)	All $m \geq 4$	2,348	1	776	1,3,1,3,13 (locks Ω-loop W_{4k})
both3 (hex + both 3)	hex + $e_{0..2}, e_{m-3..m-1}$	All $m \geq 4$	2,508	1	776	1 **for all** $m = 4 \ldots 21$ (826,564 states)

Three exact mechanisms govern the transitions across these move vocabularies. First, the precondition-gate lock $\mathcal{Z}_m$ (§148, §151) arises because a four-turn hexagon rewrite at index i is conditional: it requires the full substring 1001 or 0110 (for which vanishing central-pair parity $b_{i+1} \oplus b_{i+2} = 0$ is necessary, though not sufficient) plus post-move self-avoidance. For even $m = 2h \geq 4$, the interior pair-parity vector $p(x) = (b_2 \oplus b_3,\ b_4 \oplus b_5,\ \ldots,\ b_{m-4} \oplus b_{m-3})$ is left untouched by the end moves in both2, preserved by every even-aligned hexagon flip, and blocks every odd-aligned hexagon flip when all $p_j = 1$. Consequently, the

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zigzag family $\mathcal{Z}_m = \{x \in V_m : p(x) = (1, \ldots, 1)\}$ of cardinality $2^{(m+4)/2}$ (16,32,64,128,256,512 for $m = 4 \ldots 14$) is 100% self-avoiding, forms a single closed mirror-stable component under both2, and is completely contact-free for every even $m \geq 6$ (at $m = 4$, 0000 closes five edges of a hexagon).

Second, two-hexagon 10-cycle exchanges and the Ω-loop islands W_{4k} (§148, §149, §155) emerge because the honeycomb graph has girth 6 and no 4- or 8-cycles, making the replacement of a 5-edge arc with its complementary 5-edge arc around two adjacent hexagons (10-cycle, 4 moving beads) the next geometrically distinct local exchange. Strictly interior 10-cycle windows ($1 \leq i \leq m - 5$, acting by the 6-bit masks 101101 and 110011) produce zero $\mathcal{Z}_m$-exit edges across $m = 6 \ldots 14$, whereas the back-boundary window $i = m - 4$ (5-bit suffix masks 10110 and 11001) supplies all $2^{m/2}$ $\mathcal{Z}_m$-exit edges (8, 16, 32, 64, 128), connecting 0000100010 to 1000010010 in 5 steps and 0101010101 ∈ $\mathcal{Z}_{10}$ to 0111100101 in 7 steps. Restoring frame invariance at the front boundary $i = 0$ (5-bit prefix masks 01101 and 10011) reconnects $m = 10$ from 3 components (766, 5, 5) into 1 component (776 states), leaving at $m = 8$ and $m = 12$ only the two 4-state alternating Ω-loop islands W_{4k} and $C(W_{4k})$ built around 1111 ‖ 0000 ‖ ⋯, where adjacent neck vertices block both $e(1)$ and 10-cycle replacements.

Third, steric jamming under both3 ($m = 22 \ldots 26$, §150, §151, §155) is delayed until width 22: supplying three controllable end turns at both ends (both3) connects 100% of V_m into a single component for all $m = 4 \ldots 21$ (826,564 states at $m = 21$). At $m = 22$ ($|V_{22}| = 1{,}551{,}488$), the first both3-isolated states appear as a single 4-word Klein orbit generated by the Yin-Yang clasp $w_{22} = \text{\texttt{0001111010110110111100}}$. Across $m = 22 \ldots 26$, both3 leaves 4,12,48,162,430 states outside the giant component; adding symmetric 10-cycle exchanges unlocks w_{22} at $i = 0$ (or in-plane via the 4-bead pivot $e(3)$ with certified clearance 0.997091), leaving 0,4,4,10,28 residual planar-locked states (46 total across $m = 23 \ldots 26$, forming 7 symmetry orbits at $m = 26$).

5. Leaving the Plane: Synchronous Control Tori, Crankshaft Closure, and the Gray Decompiler

5.1 Rigid Half-Turn Convexity and Synchronous Two-Angle Execution on the 46 Planar Locks

The 46 residual conformations at $m = 23 \ldots 26$ are locked only because the chain is constrained to move flat in $\mathbb{R}^2$. In the binary mask algebra, every single-turn flip e_j factors into two consecutive suffix or prefix complement masks (8,255 width-position pairs verified through $m = 128$):

$$\boxed{e_j = D_j \oplus D_{j+1} = Q_{j-1} \oplus Q_j, \qquad D_j(x) = x \oplus (2^{m-j} - 1).}$$

Geometrically, a suffix or prefix complement reflects the moving subchain across bond axis j, which executes in three dimensions as a rigid 180° ($SO(3)$) bond-axis rotation R_π preserving all bond lengths and 120° valence angles at every intermediate angle $\theta \in [0, \pi]$. For any planar point p rotated by R_θ about an axis in the plane and any fixed planar point q, axial-transverse decomposition yields the exact convexity identity (§152, seal 997412fo…):

$$\boxed{\| R_\theta p - q \|^2 = \frac{1 + \cos\theta}{2} \| p - q \|^2 + \frac{1 - \cos\theta}{2} \| R_\pi p - q \|^2 \Longrightarrow \min_{0 \leq \theta \leq \pi} d(R_\theta E, F) = \min\{d(E, F),\, d(R_\pi E, F)\}.}$$

Thus a single rigid half-turn D_k between two valid planar conformations is guaranteed collision-free throughout its entire 3D sweep with minimum non-incident bond clearance 1.0. Allowing all valid single-D_k

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half-turns connects 100% of V_m into 1 single connected component at every tested width $m = 2 \ldots 12$ ($14{,}728$ valid edges at $m = 12$), even though the reflected Gray cycle order passes through self-intersecting states for $m \geq 5$ (§162, seal a185394f…).

To execute a localized 4-bead end gate e_j ($j \in \{3, m-4\}$) across the 46 residual locks (68 endpoint-valid gates total: 4 at $m = 23$, 8 at $m = 24$, 20 at $m = 25$, and 36 at $m = 26$), the compiler compares serial and synchronous execution schedules on the periodic two-torsion control torus $(\phi_1, \phi_2) \in [0,2\pi)^2$ (§153, §155, seals 6ec4e001…, a88c0432…). Under the serial schedule $(0,0) \to (\pi, 0) \to (\pi, \pi)$, the chain must pass through the planar intermediate $x \oplus D_j$ or $x \oplus D_{j+1}$; across the 68 gates, 56 have at least one valid planar intermediate (28 of 36 at $m = 26$, offering 44 serial routes), freeing every one of the 46 locked words via at least one end, whereas 12 gates (8 at $m = 26$) have both planar corners blocked by self-intersection. Under the synchronous schedule $(\phi_1, \phi_2) = (\theta, \sigma\theta)$ ($\sigma \in \{+1, -1\}$), rotating both adjacent bonds simultaneously turns the 4-bead end block rigidly about the pivot vertex at exact angular speed $\sqrt{2+\sigma}$ while keeping all bond lengths and $120°$ angles fixed. Across all 68 gates, the synchronous path bypasses the blocked planar corners in 3D, achieving a certified whole-sweep clearance of ≥ 0.977328 bond lengths, attained on the hardest witness $x^* = \text{\texttt{0101111010110101111010111110}}$ where both planar arcs collide, gate 3 counter-rotating is certified at 0.984885, and gate 22 co-rotating is certified at 0.977328 (while the coarse 14-Lipschitz bound on the 90-step grid gives $1 - 14\pi/180 = 0.755654$).

5.2 Strictly Interior Crankshaft Closure, the Face-Boundary Compiler, and Sum-of-Squares Rigidity

When a two-torsion move $e_j = D_j \oplus D_{j+1}$ is applied to an interior bit ($j = 5$ of 0101010101 at $m = 10$), the two adjacent bond axes are noncollinear ($120°$ apart); since two nontrivial proper rotations about noncollinear axes cannot compose to the identity, the move rotates the downstream exterior by $+120°$ in the plane (DOWNSTREAM_RIGID_ROTATION_ERR = 3.553e-15, §154, seal f8ee06df…). By contrast, in every 4-turn hexagon rewrite 1001 ↔ 0110, the entering bond (p_i, p_{i+1}) and exiting bond (p_{i+4}, p_{i+5}) are collinear with the interior chord $p_{i+1}p_{i+4}$. Synchronized opposing torsions $(\phi_L, \phi_R) = (\theta, -\theta)$ about those two collinear exterior bonds cancel identically on the distal chain for all $\theta \in [0, \pi]$, rotating only the two internal beads p_{i+2}, p_{i+3} around the chord with zero exterior displacement and exact endpoint-convexity clearance, verified across $10{,}591$ even-width edges for $m = 4 \ldots 14$ and $5{,}016$ odd-width edges for $m = 5 \ldots 13$, totaling $15{,}607$ edges (seals 21b9add8…, 32bd2f15…).

Grouping turn windows by their $SE(2)/SE(3)$ running product $G_k = G_0 g_1 \cdots g_k$—or equivalently constructing the outer perimeter $L = 6h - 2s$ of h connected hexagons sharing s internal edges (§158, §160)—generates the complete multi-hexagon exchange hierarchy without lattice search:

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Cluster Perimeter L=6h−2sL = 6h - 2sL=6h−2s	**Turn Width ℓ\ell ℓ**	**Moving Beads mb=k−1 m_b = k - 1mb =k−1**	**Valid Cuts**	**XOR Mask Classes**	**Exterior Axis Geometry & Motion Type**	**Planar Jacobian (Eqs,Vars,Rank,Nullity)(\text{Eqs}, \text{Vars}, \text{Rank}, \text{Nullity})(Eqs,Vars,Rank,Nullity)**	**Generic 3D Freedom k−4k - 4k−4**
6-Cycle $(h = 1, s = 0)$	4	$2\ (k = 3)$	$1/1$	1111	Collinear flanks; **rigid** $180°$ **crankshaft** (locally rigid-only)	$(7,6,4,2)$	0 (1D rigid circle)
10-Cycle $(h = 2, s = 1)$	6	$4\ (k = 5)$	$4/5$	101101 (cuts $0,3$) 110011 (cuts $1,2$)	Parallel offset flanks $(\Delta y_{\perp} = \sqrt{3}, \sqrt{3}/2)$; **1D loop pucker** (2 & 5 real exit tangents)	$(11,12,8,4)$	1 (1D curve)
12-Cycle $(h = 3, s = 3,$	7	$5\ (k = 6)$	$6/6$	111111 (cuts $0,2,4$)	111111: Collinear **rigid crankshaft**	$(13,15,10,5)$	0 (rigid circle)

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Cluster Perimeter L=6h−2sL = 6h - 2sL=6h−2s	**Turn Width ℓ\ellℓ**	**Moving Beads mb=k−1m_b = k - 1mb =k−1**	**Valid Cuts**	**XOR Mask Classes**	**Exterior Axis Geometry & Motion Type**	**Planar Jacobian (Eqs,Vars,Rank,Nullity)(\text{Eqs}, \text{Vars}, \text{Rank}, \text{Nullity})(Eqs,Vars, Rank,Nullity)**	**Generic 3D Freedom k−4k - 4k−4**
triangle)				1010101 (cuts 1,3,5)	(SOS rigid-only) 1010101: Non-parallel **2D pucker**		2 (2D surface)
14-Cycle ($h = 3, s = 2$, chain)	8	6 ($k = 7$)	5 /7	11111111 (cut 5) 10100101 (cuts 0,3) 11000011 (cuts 1,2)	11111111: Collinear crankshaft + 3D flex 10100101, 11000011: Parallel **3D pucker**	(15,18,12,6)	3 (3D manifold)

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At any planar honeycomb configuration, all out-of-plane z-columns of the Cartesian constraint Jacobian $J = DF(X_0)$ vanish identically, creating a singular bifurcation governed at second order by $JY + q(z) = 0$, or $\ell_r^{\mathsf{T}} q(z) = 0$ for every left-null vector $\ell_r \in \ker J^{\mathsf{T}}$ (§160, §162, seals 831a9024…, 99ab12a3…). For the 12-cycle collinear crankshaft 0101111 ($\mathrm{rank} J = 10$, $\dim\ker J^{\mathsf{T}} = 3$), decomposing $z = r(1,1,2,2,1) + (0, u_2, u_3, u_4, u_5)$ modulo the rigid crankshaft mode r cancels r from all three quadratic forms Q_1, Q_2, Q_3 and yields an exact positive-definite sum-of-squares (SOS) certificate (§162):

$$\boxed{2(2Q_1 + 9Q_2 + 13Q_3) = (u_2 + u_3 - u_4)^2 + 6\left(u_3 - u_4 + \frac{1}{3}u_5\right)^2 + 9\left(u_4 - \frac{5}{3}u_5\right)^2 + \frac{13}{3}u_5^2 = 0,}$$

with leading principal minors $[1,6,54,234]$ and an independent certificate $10Q_1 + 49Q_2 + 70Q_3$ having minors $[13/2, 105, 19755/8, 24225]$. Thus $u_2 = u_3 = u_4 = u_5 = 0$, proving rigorously that no nonrigid pucker branch can leave the planar 12-cycle crankshaft at second order (replacing an earlier false $d^2 \le 13$ span argument, since the 5-bond zigzag reaches $d^2 = 19$). Across all 15 collinear-flank crankshaft windows up to $\ell = 10$, only $\ell = 4$ (1111) and $\ell = 7$ (1111111) are locally rigid-only; all 12 windows at $\ell = 8,9,10$ admit real transverse flexing directions.

Similarly, at the 12 residual $m = 26$ locks that admit 12-/14-cycle boundary exchanges (8 rigid half-turns and 4 pucker-only words 00001111011010110110111100, 00111101101101011011110000 and their complements, §159–§160), the planar end-window Jacobian has 14 equations in 18 coordinates, rank 12, nullity 6, left cokernel dimension 2, and internal bond-constraint Hessian spectral norm 4 (not 2). Because all 4 pucker-only words are related by J and C and sit at a free chain terminus ($i = 0$ or $i = 20$), unpinning the free end and sweeping its 5 active torsions via forward NeRF kinematics preserves exact bond lengths and 120° angles throughout while maintaining whole-chain clearance ≥ 0.965 bond lengths, complementing the 200-step projected bridges and 579-/484-step continuations with piecewise-linear bounds 0.959451 and 0.964478. Finally, §161 (seal c0b34365…) establishes that an intrachain H–H contact is a single scalar distance equation $\| p_b(\phi) - p_a(\phi) \|^2 - d_{ab}^2 = 0$, removing 1 generic degree of freedom versus 3 for a point weld and 6 for a full $SE(3)$ pose dock. On 1000010010, rotating the tail about $p_2 \to p_3$ gives $\| p_6(\theta) - p_1 \|^2 = 4 - 3\cos\theta$ ($F'(0) = 0, F''(0) = 3$); across all 1,598 valid chains up to $m = 10$, all 226 planar contacts are first-order flat ($\nabla F = 0$), splitting at second order into 220 hexagon contacts ($b = a + 5$) with positive-definite rank-3 Hessians and 6 nine-bond contacts ($b = a + 9$) with rank-7 indefinite Hessians.

6. The Integer Boundary Operator and the Three-Dimensional Diamond-Lattice Compiler

6.1 The Shared Coboundary Operator Across Binary and Real Domains

The transition from discrete ring operations to physical force laws is anchored in §163 (seal 046fcc9f…) by recognizing that the binary difference reader δ and the continuum mechanical strain operator are two realizations of the same universal integer coboundary row:

$$\boxed{B = \begin{pmatrix} -1 & 1 \end{pmatrix}, \qquad Bb \pmod 2 = b_1 \oplus b_0 = \delta(b), \qquad Bx = x_1 - x_0 = d(x).}$$

Because no nontrivial additive group homomorphism $\mathbb{F}_2 \to \mathbb{R}$ exists ($2f(1) = f(0) = 0 \Longrightarrow f(1) = 0$), the correspondence does not embed binary values into real numbers; rather, both domains implement the same integral incidence relation, and reversing either readout requires a held reference ($\bar{x} = (x_0 + x_1)/2$ over $\mathbb{R}$,

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or b_0 over $\mathbb{F}_2$). With the two-site exchange involution $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, any symmetric quadratic two-site potential $U(x) = \frac{1}{2}x^{\mathsf{T}}Kx$ that is invariant under uniform translation ($K(1,1)^{\mathsf{T}} = 0$) is uniquely forced to take the form:

$$\boxed{K = \kappa B^{\mathsf{T}}B = \kappa(I - S), \qquad P_{\pm} = \frac{I \pm S}{2}, \qquad \omega_{+}^2 = 0, \qquad \omega_{-}^2 = \frac{2\kappa}{m}.}$$

Extending B to a general graph incidence matrix B_G yields $K = B_G^{\mathsf{T}}WB_G$, fixing the dimensionless normal-mode frequency ratios from connectivity alone ($\omega^2 \in \{0, \kappa/m, 3\kappa/m\}$ with ratio $\sqrt{3}$ for a 3-node path, versus $\{0,3\kappa/m, 3\kappa/m\}$ with ratio 1 for a 3-node triangle). In phase space $y = (u, v)^{\mathsf{T}}$, combining S with the sign reader $Z = \mathrm{diag}(1, -1)$ forms the symplectic generator $J = ZS$ ($J^2 = -I, SZ = -ZS$) and damping projector $Q = (I - Z)/2$, producing the universal damped linear response $\dot{y} = (\omega J - \gamma Q)y$ that governs both mechanical oscillators ($u = \sqrt{\kappa}x, v = \sqrt{m}\dot{x}$) and series RLC circuits ($u = q/\sqrt{C}, v = \sqrt{L}i$). At the first-order-flat §161 contact $c(\theta) = 3(1 - \cos\theta)$, a harmonic penalty $U = \frac{\eta}{2}c(\theta)^2$ vanishes at second order and generates a purely quartic nonlinear restoring potential $U(\theta) = \frac{9\eta}{8}\theta^4 + O(\theta^6)$.

6.2 Three-Dimensional Diamond-Lattice Exchanges and the Two-Sheeted Mirror Covering

Moving from the planar honeycomb chain to a realistic three-dimensional tetrahedral backbone replaces the two planar turns with the four tetrahedral bond directions $b \in \{\pm 1\}^3$ on the diamond lattice ($\| b \|^2 = 3$, consecutive forward dot product $b_{j-1} \cdot b_j = +1$, tetrahedral valence angle $\arccos(-1/3) \approx 109.47°$) and three rotational-isomeric torsion states: trans (180°), gauche$^+$ (+60°), and gauche$^-$ (−60°). Grouping ℓ-torsion windows ($k = \ell - 2$ moving arc bonds, $m_b = k - 1$ moving beads) by their $SE(3)$ running product modulo the S_3 symmetry group of the initial bond $b_1 = (1,1,1)$ classifies all local tetrahedral exchanges without search.

At $k = 3$ ($\ell = 5$, 2 moving beads closing a 6-bond diamond chair hexagon), 4 intrinsic S_3-orbits appear, including the parallel-flank t g$^+$t ↔ t g$^-$t Boyer/Helfand move and the g$^+$g$^+$g$^+$ ↔ g$^-$g$^-$g$^-$ chair flip. Unlike the planar honeycomb hexagon where $\cos 60° = 1/2$ makes the flanking bonds collinear, the tetrahedral angle $\cos\theta = -1/3$ offsets the flanking bonds laterally in all 4 orbits, giving the constraint Jacobian full rank 6 in 6 variables (nullity 0). Thus Boyer's three-bond exchange is kinematically isolated under strict fixed-exterior constraints. At $k = 5$ ($\ell = 7$, 4 moving beads, 11 length and angle constraints in 12 Cartesian coordinates, generic 6R linkage freedom $k - 4 = 1$), there are 38 intrinsic S_3-orbits. As established by the diamond-lattice audit correction, 16 of the 38 seven-bond ($\ell = 7, k = 5$) exchange classes connect numerically under exact bond-length and tetrahedral-angle constraints, correcting the preliminary 13/38 count. In particular, Schatzki's five-bond crankshaft (Orbit 13) has collinear exterior flanks along $x = y = z$ and connects via an exact 120° proper $SO(3)$ rotation ($\det = +1$), whereas improper mirror-enantiomer pairs ($\det = -1$, such as Orbit 14 exchanged by the reflection $R(x, y, z) = (z, y, x)$) lie on disjoint 1D circles when $p_6 = (4,4,4)$ is held fixed.

To test whether the mirror-related Orbit 14 endpoints A and $B = RA$ can connect when one boundary bond is released (§164–§165), the compiler extends the 5-bond arc by the unique reflection-invariant diamond bond $b_{\text{new}} = p_8 - p_7 = (1, -1, 1)$, holding $p_0 = (0,0,0), p_1 = (1,1,1), p_7 = (5,5,5), p_8 = (6,4,6)$ fixed while

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$p_2 \ldots p_6$ move in $\mathcal{M} \subset \mathbb{R}^{15}$. This promotes the exact rational Jacobian from rank $11/12$ (nullity 1) to rank $13/15$ (nullity 2) and unlocks the transverse velocity $\dot{p}_6 = (-1,0,1)$, although a 50-step straight-line chord projection stalls 1.4694 units from B.

Rather than guessing trajectories, §165 (seal ac38e442…) determines the topology of the mirror quotient $\pi: \mathcal{M} \to \mathcal{M}/\langle R \rangle$ by running the boundary constraints in reverse on the mirror-fixed plane $x = z$. In the orthonormal basis $u = (1,1,1)/\sqrt{3}$ and $v = (1,-2,1)/\sqrt{6}$ of $x = z$, each planar bond steps by angle $\pm\alpha$ where $\cos\alpha = 1/3$, corresponding to the algebraic multiplier $z = e^{i\alpha} = \frac{1+2i\sqrt{2}}{3} \in \mathbb{Q}(i\sqrt{2})$. Because $2\cos\alpha = 2/3$ is not an algebraic integer, z is not a root of unity, fixing the terminal heading uniquely at $a_7 = 1$. Any mirror-fixed configuration ($RP = P$) must satisfy the exact sum constraint:

$$\boxed{\sum_{j=1}^{6} z^{a_j} = 4, \qquad a_0 = 0, \qquad a_7 = 1, \qquad a_j - a_{j-1} \in \{-1, +1\}.}$$

Propagating the residual backward from $(a_7, \Delta) = (1,4)$ in $\mathbb{Z}[\sqrt{-2}, 1/3]$ across layers $j = 7 \ldots 1$ (2, 4, 8, 15, 28, 50, 90 distinct algebraic states out of 128 binary words) filters to $\binom{7}{3} = 35$ boundary-compatible heading histories and proves that zero reflection-fixed configurations exist anywhere in $\mathcal{M}$:

$$\boxed{\min_{35\ \text{histories}} |\Delta_{\text{norm}}|^2 = \frac{8}{9} \quad \left(\text{attained 3 times each at } \pm\frac{2i\sqrt{2}}{3}\right) \Longrightarrow \min_{P=RP} \| p_7 - p_1 - (4,4,4) \|^2 = 3 \times \frac{8}{9} = \frac{8}{3}.}$$

Because $\mathrm{Fix}(R) \cap \mathcal{M} = \emptyset$, the $\mathbb{Z}_2$ reflection R acts freely on $\mathcal{M}$, making $\pi: \mathcal{M} \to \mathcal{M}/\langle R \rangle$ an unbranched two-sheeted covering with monodromy homomorphism $\mu: \pi_1(\mathcal{M}/\langle R \rangle, \pi(A)) \to \mathbb{Z}_2$. A physical path $A \to B = RA$ exists if and only if the quotient admits a loop with odd monodromy $\mu = 1$—the exact continuous counterpart of the ring's twisted clock $\tau^n = C_n, \tau^{2n} = I$, where $q(\tau^n x) = q(Cx) = q(x)$ closes in the quotient while switching sheets in the cover.

7. Quantitative Matter Readout: Symmetry Reduction and the Inverse Rank Problem in Polyethylene

7.1 Reduction of the 72-State Tetrahedral Transfer Model to Four Scalar Coordinates

The last unresolved item in the ledger through §163 was the first quantitative matter readout: deriving a bulk polymeric observable directly from the tetrahedral shape operations without stochastic sampling or parameter fitting. In the rotational-isomeric-state (RIS) formulation of unperturbed polyethylene (polymethylene) chains, each C–C bond lies along one of the 8 signed tetrahedral diamond vectors $\{\pm 1\}^3$ ($\| b \| = \sqrt{3}$), yielding 24 oriented two-bond frames (a, b) satisfying $a \cdot b = 1$ ($\cos\theta_{\text{C-C-C}} = 1/3$, valence angle $109.47°$). Given an oriented bond pair (a, b), the next bond d is selected deterministically by one of the three torsion states $tw \in \{\text{trans}, \text{gauche}^+, \text{gauche}^-\}$: the trans ($180°$) continuation sets $d = a$ (preserving the (a, b) plane), whereas the gauche$^{\pm}$ ($\pm 60°$) continuations select the two remaining tetrahedral vectors $d \neq a$ satisfying $b \cdot d = 1$, distinguished by the sign of the oriented bond-triple determinant $\det([a, b, d]) \gtrless 0$.

Tracking the 24 oriented bond pairs together with the 3 preceding torsion states produces a 72-state transfer operator $T_{72\times 72}$. The local relative Boltzmann weight matrix U between consecutive torsion states

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(trans, gauche$^+$, gauche$^-$) is parameterized by two physical energies—the trans/gauche energy difference E_g ($\sigma = e^{-E_g/RT}$) and the neighboring opposite-gauche (g$^\pm$g$^\mp$ pentane-interference) penalty E_p ($\omega = e^{-E_p/RT}$):

$$\boxed{U = \begin{pmatrix} 1 & \sigma & \sigma \\ 1 & \sigma & \sigma\omega \\ 1 & \sigma\omega & \sigma \end{pmatrix}, \qquad P_{ij} = \frac{U_{ij} h_j}{\lambda_{\max} h_i},}$$

where $(\lambda_{\max}, h)$ is the Perron–Frobenius eigenpair of U and P is the Doob-transformed stochastic transition matrix of the infinite Gibbs chain with stationary distribution $\pi = (\pi_t, \pi_{g^+}, \pi_{g^-})$ ($\pi P = \pi$, $\pi_{g^+} = \pi_{g^-}$).

In the local orthogonal bond frame $(a, b, c = a \times b)$, advancing one step to $(b, d, b \times d)$ transforms vectors by the exact tetrahedral matrices:

$$M_{\text{trans}} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad M_{\text{gauche}^+} = \begin{pmatrix} 0 & -1/2 & 3/2 \\ 1 & 1/2 & -1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}, \qquad M_{\text{gauche}^-} = \begin{pmatrix} 0 & -1/2 & -3/2 \\ 1 & 1/2 & 1/2 \\ 0 & -1/2 & 1/2 \end{pmatrix}.$$

While solving $(I_9 - (P \otimes I_3)\text{diag}(M_t, M_{g^+}, M_{g^-}))x = \boldsymbol{e}_2$ already reduces the 72-state spatial system to 9 frame-correlation unknowns, §167 establishes a further exact symmetry reduction to just four scalar coordinates: because the local reflection R exchanges gauche$^+$ ↔ gauche$^-$ while inverting the cross-product normal $c = a \times b \mapsto -c$, the gauche$^+$ and gauche$^-$ longitudinal components (x_1, x_2) are identical and their binormal components x_3 are equal and opposite, while trans has zero binormal expectation by reflection symmetry. Consequently, the entire 72-state chain-dimension problem collapses to four scalar unknowns, agreeing numerically with the full 72-state linear solver to $\approx 10^{-15}$ (1.33e-15 to 5.33e-15).

Solving the 4-coordinate system symbolically yields two exact closed-form theorems before any energy penalty is fitted. First, in the unweighted tetrahedral baseline ($\sigma = 1, \omega = 1$), where all three torsion states are equally weighted, the infinite-chain characteristic ratio $C_\infty = \lim_{n\to\infty} \frac{\langle r^2 \rangle_0}{n\ell^2} = 1 + 2\sum_{k=1}^{\infty} c\,(k)$ evaluates identically to:

$$\boxed{C_\infty = 2, \qquad c(k) = \frac{\langle b_0 \cdot b_k \rangle}{3} = 3^{-k}.}$$

Second, in the independent-torsion Boltzmann regime ($\omega = 1$, arbitrary σ), where a trans/gauche Boltzmann factor σ is present without a neighboring-gauche penalty, the symbolic 4-coordinate reduction collapses to:

$$\boxed{C_\infty = \frac{2(2+\sigma)}{3\sigma}.}$$

7.2 Physical Readouts at 413 K and the Reverse-Gap Rank Theorem

Evaluating the 4-coordinate and 72-state readers using the literature RIS energies for polyethylene ($E_g = 2100$ J/mol ≈ 0.5 kcal/mol and pentane-interference penalty $E_p = 8400$ J/mol ≈ 2.0 kcal/mol, with $R = 8.314462618$ J/(mol · K)) without parameter adjustment produces the following exact readouts:

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Polyethylene RIS Model Regime	Temp TTT (K)	Boltzmann σ=e−Eg/RT\sigma = e^{-E_g/RT}σ=e−Eg/RT	Penalty ω=e−Ep/RT\omega = e^{-E_p/RT}ω=e−Ep/RT	Characteristic Ratio C∞C_\inftyC∞	4-Coord vs. 72-State ‖Δ|\Delta‖Δ	Bond Correlations c(1),c(2),c(3),c(4)c(1), c(2), c(3), c(4)c(1),c(2),c(3),c(4)
All torsions equally weighted	—	1.00000000	1.000000000	2.0000 (2.0000000)	1.33×10^{-15}	0.333333,0.111111,0.03
Trans/gauche energy E_g only	413 K	0.54250696	1.000000000	3.1244 (3.1243923)	8.88×10^{-16}	0.333333,0.306151,0.15
Both E_g and E_p included	413 K	0.54250696	0.086620620	6.2297 (6.2296763)	5.33×10^{-1}	0.333333,0.462219,0.31
Opposite-gauche forbidden ($\omega \to 0$)	413 K	0.54250696	0.000000000	6.7153	$\approx 10^{-15}$	Asymptotic $E_p \to \infty$ ceiling at fixed E_g
Room-temperature referen	300 K	0.43088825	0.034471377	7.2007 (7.2006503)	3.55×10^{-15}	0.333333,0.524036,0.32

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Polyethylene RIS Model Regime	Temp TTT (K)	Boltzmann σ=e−Eg/RT\sigma = e^{-E_g/RT}σ=e−Eg/RT	Penalty ω=e−Ep/RT\omega = e^{-E_p/RT}ω=e−Ep/RT	Characteristic Ratio C∞C_\inftyC∞	4-Coord vs. 72-State ‖Δ|\Delta‖Δ	Bond Correlations c(1),c(2),c(3),c(4)c(1), c(2), c(3), c(4)c(1),c(2),c(3),c(4)
ce (E_g, E_p)						
SANS / Explicit-Atom MD Benchmark	413 K	—	—	7.8 ± 0.4 (MD ≈ 7.9)	—	Requires relaxed $\theta_{\text{C-C-C}} \approx 112°, \phi_g \approx 113°$

A 1,001-point numerical sweep of the neighboring-gauche penalty $\omega \in [0,1]$ at fixed 413 K trans/gauche weight $\sigma = 0.542506969$ confirms that C_∞ increases monotonically as ω decreases from 1 (3.1244) to 0 (6.7153). This exposes a decisive physical finding: even when opposite-gauche ($g^{\pm}g^{\mp}$) transitions are completely forbidden ($E_p \to \infty$), the idealized tetrahedral lattice reader ($\theta = 109.47°, \phi = 180°, \pm 60°$) reaches at most $C_\infty = 6.7153$, matching the older viscosity estimate of ≈ 6.8, whereas the small-angle neutron scattering (SANS) central estimate is ≈ 7.8 and explicit-atom molecular dynamics gives ≈ 7.9 at 413 K (which Han, Jaffe, and Yoon demonstrated requires opened valence angles $\approx 112°$ and shifted gauche minima $\phi_g \approx 113°$). The gap between 6.2297 (or 6.7153) and 7.8 cannot be dismissed as floating-point error or an approximation in the 72-state transfer calculation; it is an exact property of the declared ideal-geometry model.

Finally, §167 formulates the reverse gap—inferring the effective microscopic parameters (E_g, E_p) from macroscopic measurements—as a rigorous Jacobian rank problem. Because the model depends on two effective energy parameters (E_g, E_p), measuring C_∞ alone supplies only one scalar equation for two unknowns, creating a one-dimensional level curve of indistinguishable parameter pairs. At the 413 K operating point ($E_g = 2100$ J/mol, $E_p = 8400$ J/mol), implicit differentiation along constant $C_\infty = 6.2297$ yields the local tradeoff slope:

$$\left.\frac{dE_p}{dE_g}\right|_{C_\infty} \approx -5.8812.$$

The shape channel identifies the exact missing reader required to invert the map: a second independent observable on the same 3-state torsion distribution, such as the equilibrium trans-state population $p_t =$

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π_{trans}, which the model predicts to be $p_t \approx 0.596664$ at 413 K. Evaluating the two-observable sensitivity Jacobian $J = \frac{\partial(C_\infty, p_t)}{\partial(\text{parameters})}$ at the 413 K operating point yields:

$$\boxed{J = \begin{pmatrix} 0.790254 & 0.134369 \\ 0.052463 & 0.003102 \end{pmatrix}, \qquad \det J \approx -0.004598 \neq 0, \qquad \operatorname{rank} J = 2.}$$

Because J has full rank 2, pairing C_∞ with the equilibrium trans population removes the local algebraic ambiguity and makes the parameter-to-observable map locally invertible, although the inverse remains poorly conditioned due to the small determinant $\det J \approx -0.004598$. Thus the tetrahedral shape operations supply the transfer law, its exact 72-to-4 symmetry reduction, and the required two-observable rank structure without any further search over ring assignments; what remains OPEN is linking an independent experimental readout of the trans population (for instance, via Raman or ^{13}C NMR spectroscopy) and relaxing the fixed tetrahedral angles to close the empirical gap at 413 K.

8. Consolidated Master Ledger and Conclusions

In strict adherence to the project audit protocol, the following master ledger records the proven theorems (STANDS), verified finite and symbolic executions (RUNS), falsified routes with their true owners (DEAD END), and prospective empirical or analytical interfaces (OPEN) across the unified architecture:

Section & Module	Mathematical Identity / Operational Claim	Verification Scope & Seal / Provenance	Status
§§135–143: Native Ring & *One Inverter*	Fixed-point-free partner $P_n(x)$; facing law $R_n[N-1-i] = P_n(R_n[i])$; rank $r_n(p)$ with $T_n(p)$ clamp; twisted clock $\tau^n = C_n, \tau^{2n} = I$; center reader $r_n(x) = x_{n/2-1} \oplus x_{n/2}$; odd-lane locks $\Lambda_q(u)$.	Exhaustive $n = 2 \ldots 20$ (2,097,148 words); scalar queries to $n = 1{,}024$; 51/51 checks under seal 00981594….	**STANDS / RUNS**
§§135–143: Square-Root Bridge & Rungs	$PTP = T^{-1}$, $\langle T, T_C \rangle_F = 0$, $B = T_C T^{-1} = S^2$; LEN=6 detachment [32,16,16]; global C-paired rungs M and reflection $H = MC$; all-width 16-branch (4/12 ports) and 28-branch (1/27 ports) symbolic certificates.	Exhaustive $n = 2 \ldots 16$ (131,068 states); wide checks to $n = 2{,}048$; seals 441c0558…, 39c153fa…, d8221144…, 250ecd1b….	**STANDS / RUNS**

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Section & Module	Mathematical Identity / Operational Claim	Verification Scope & Seal / Provenance	Status
§§135–143: R_{64} Dual-Read & Address Jumps	Center-2 quotient ($2^{n/2}$ exceptions); palindromic fold $D(x) = x \oplus P_{64}(x)$; tagged 2-bit shift inverse; k_{out} face selector and k_{in} quotient boundary; $\Delta u = \Delta f + \Delta \varepsilon$ ($	\varepsilon	2^{32} < 0.248$).
§144: Square-Lattice PHPPHP Toy Audit	0 single/double-site catalog edges; cyclic T_8 hopping crosses 4 padding edges with leakage $\frac{49}{75}t^2$ and virtual bridges $0.49/\Lambda$; projected PKP splits into two P_3 paths (contact cap $\leq 1/2$).	Exact 8×8 matrix audit and Strang error series ($r = 4 \ldots 128$); seal of6ocd3e….	**STANDS / RUNS**
§§145–155: Honeycomb Compiler & *The Way Between*	Turn-word bijection $x \leftrightarrow (\text{face}, \delta(x))$; $C_m = R_{\pi,x}$ vs. chiral P_z; parity split $H = H_+ \oplus H_-$; both2 traps $\mathcal{Z}_m$ ($2^{(m+4)/2}$); 10-cycles exit $\mathcal{Z}_m$ at boundary; both3 connects $m = 4 \ldots 21$; $68/68$ gates certified ≥ 0.977328 in 3D.	Exhaustive $m = 4 \ldots 26$ ($	V_{26}
§§154–162: $SE(3)$ Crankshafts, SOS & Gray Ring	Collinear dual-torsion $(\theta, -\theta)$ on 15,607 hex edges (10,591 even + 5,016 odd); $L = 6h - 2s$ hierarchy; exact SOS local rigidity of 12-cycle 0101111 (minors [1,6,54,234]); $\delta(Px) = \text{rev}(\delta(x))$; valid D_k graph connected $m = 2 \ldots 12$.	Exact rational Jacobians and Sturm eliminants; 8,188 distance/partner checks; 18,432 reflection checks; seals 21b9add8…, a185394f….	**STANDS / RUNS**
§§160–161: Planar Singularity & Contact Rank	End-window planar Jacobian rank $12/18$, left cokernel 2 ($\ell_r^{\mathrm{T}} q(z) = 0$), internal bond Hessian norm 4; contact rank 1 vs. weld 3 vs. pose dock 6; $\|p_6(\theta)-p_1\|^2 = 4 - 3\cos\the$; all $ $2planarcontactshave \nabla F =$.	Exact rational SymPy audits and 1,598-chain contact Hessian census; seals 831a9024…,	**STANDS / RUNS**

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Section & Module	Mathematical Identity / Operational Claim	Verification Scope & Seal / Provenance	Status
		99ab12a3…, cob34365….	
§§163–166: Boundary Operator & Diamond Lattice	Shared coboundary $B = (-1,1)$ over $\mathbb{F}_2$ and $\mathbb{R}$, $K = \kappa(I - S)$, quartic contact $\frac{9\eta}{8}\theta^4$; Boyer 3-bond move isolated (rank $6/6$); Schatzki $120°$ crankshaft and **16/38** 5-bond classes connect; $\mathrm{Fix}(R) \cap \mathcal{M} = \emptyset$ (min gap$^2 = 8/3$).	Exact symbolic and $\mathbb{Q}(i\sqrt{2})$ backward transfer across 35 histories; seals 046fcc9f…, ac38e442….	**STANDS / RUNS**
§167: Polyethylene 4-Coordinate Matter Reader	Reflection symmetry reduces 72-state RIS transfer model to 4 scalar coordinates (matching 72-state solver to 10^{-15}); $C_\infty = 2$ (unweighted) and $\frac{2(2+\sigma)}{3\sigma}$ ($\omega = 1$); 413 K readouts $2.0000, 3.1244, 6.2297, 6.7153$ ($\omega \to 0$); Rank-2 Jacobian ($\det J \approx -0.004598$) with $p_t \approx 0.596664$.	Symbolic reduction, 72-state and 4-coordinate numerical solvers, 1,001-point ω sweep, and 2×2 Jacobian audit.	**STANDS / RUNS**
Logged Corrections & Falsified Routes	(1) Differential map $d(x)$ as simple Q_3 graph cover (vertex quotient only); (2) $7/9 \to 4/12$ ports on 16-branch ($n \geq 7$) and 6-phase failure of naive middle decoder; (3) $d^2 \leq 13$ 5-bond span bound ($d^2 = 19$, replaced by SOS certificate); (4) 6-DOF clamp for single point contact (rank 1).	Explicitly routed to their true mathematical owners in §§5, 13, 161, 162.	**DEAD END** (routed)
Prospective Empirical & Analytical Links	(1) Certified continuous manifold enclosures for strictly interior fixed-exterior nonrigid puckers; (2) physical steric radii, solvent coupling, and kinetic rates; (3) linking an independent experimental readout of p_t at 413 K and relaxed valence/torsion angles ($\theta \approx$	Typed as exact missing providers rather than filled by convention.	**OPEN**

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Section & Module	Mathematical Identity / Operational Claim	Verification Scope & Seal / Provenance	Status
	$112°, \phi_g \approx 113°$) to close the 6.7153 → 7.8 SANS gap.		

Synthesis of Conclusions

The completed §§135–167 trajectory demonstrates that the relational structure of the in-between serves as the governing invariant across discrete and continuous domains. The bridge from discrete binary rings to continuous mechanical and polymeric matter does not assign particles to bits or import hardware metaphors into physics. Instead, the integer coboundary row $B = (-1 \quad 1)$, the reflection involutions (J, C, P, R), and the running group product $G_n = G_0 g_1 \cdots g_n$ define an invariant relational grammar. Evaluated over $\mathbb{F}_2$, B acts as the dihedral shape reader $\delta(x)$ of the ring; evaluated over $\mathbb{R}$, $-B^{\mathsf{T}} \kappa B$ yields the unique translation-invariant harmonic force law; and evaluated on a planar contact with vanishing first derivative, $\frac{\eta}{2} c(\theta)^2$ generates a quartic restoring mode $\frac{9\eta}{8}\theta^4$.

Simultaneously, schedule and dimensionality resolve apparent topological locks without ad hoc state-space shortcuts. Across both the two-dimensional honeycomb lattice and the three-dimensional diamond lattice, configurations that appear permanently trapped under sequential planar rewrites or rigid 5-bond boundaries unlock when the compiler expands into the continuous control torus $(\phi_1, \phi_2) = (\theta, \sigma\theta)$ or releases a single boundary bond to traverse a two-sheeted reflection cover $\mathcal{M}/\langle R \rangle$. Conversely, when a configuration is genuinely rigid—such as the 12-cycle collinear crankshaft 0101111, Boyer's 3-bond diamond exchange ($k = 3$), or the mirror-fixed plane $x = z$ in §165—the compiler certifies its rigidity algebraically via exact sum-of-squares forms or $\mathbb{Q}(i\sqrt{2})$ backward transfer residuals rather than relying on failed numerical searches.

Finally, symmetry reduction transforms macroscopic matter readout into a finite rank problem. In §167, applying the local gauche$^+$ ↔ gauche$^-$ reflection quotient collapses the 72-state tetrahedral polyethylene transfer model to four scalar unknowns, yielding the exact closed forms $C_\infty = 2$ and $C_\infty = \frac{2(2+\sigma)}{3\sigma}$ and proving that the ideal tetrahedral geometry caps C_∞ at 6.7153 at 413 K even when opposite-gauche transitions are forbidden. By demonstrating that (C_∞, p_t) possesses a rank-2 Jacobian ($\det J \approx -0.004598$), the shape channel reduces the inverse material-parameter problem to a well-posed, two-observable algebraic closure.

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